

PEDRO RAMOS

Bachelor's Student in Electromechanical Engineering

Leuven, Belgium

📞 +352 621 750 018 ✉ ramospedro1810@gmail.com 🔗 linkedin.com/in/pedroramos 🌐 olympusmars.be

Profile

Bachelor's student in Electromechanical Engineering at KU Leuven specializing in robotics, automation, and autonomous systems. Co-founder of Olympus MARS, where I help develop a Mars rover for the European Rover Challenge. Experienced in ROS2, mechanical design, embedded systems, and multidisciplinary engineering projects, with a strong interest in intelligent robotic systems.

Education

KU Leuven – Group T

B.Sc. in Electromechanical Engineering

Leuven, Belgium

Sep. 2023 – Expected Sep. 2026

Lycée Classique d'Echternach

Natural Sciences – Mathematics

Echternach, Luxembourg

Sep. 2020 – July 2023

Technical Skills

Programming: Python • Java • MATLAB • SQL

Robotics: ROS2 • OpenCV • Embedded Systems

Automation: PLC Programming • Siemens TIA Portal

CAD & Simulation : Siemens NX • MATLAB/Simulink

Software & Tools : Git • HTML/CSS

Engineering Projects

Olympus MARS – Mars Rover Development | Co-Founder

2026

- Founded a student engineering team developing an autonomous Mars rover for the European Rover Challenge.
- Built the team's recruitment pipeline and structure from zero, growing the team to 20 members.
- Defined the architecture and technical requirements for the rover's robotic arm subsystem for the European Rover Challenge.
- Collaborated with students from multiple engineering disciplines on a large-scale robotics project.

Autonomous Robotic Harvesting System

Sep 2025 – June 2026

- Designed a robotic harvesting solution consisting of a robotic arm mounted on an autonomous guided vehicle (AGV).
- Contributed to mechanical design, system integration, and digital twin development.
- Performed engineering calculations related to motion planning, positioning accuracy, and mechanical performance.
- Developed the system as part of an advanced mechatronics engineering project.

ROS2 Communication System

July 2026

- Developed ROS2 nodes in Python for real-time communication with embedded sensors.
- Implemented serial communication and JSON message parsing.
- Integrated ultrasonic sensor data into a ROS2 ecosystem.
- Diagnosed and resolved hardware–software communication issues.

Experience

Olympus MARS

2026 – Present

Co-Founder

Leuven, Belgium

- Founded a robotics student team focused on developing an autonomous Mars rover.
- Coordinated recruitment and external communication.
- Supported technical planning and project organization.
- Developed and deployed the Olympus MARS website using HTML/CSS, including domain registration, hosting configuration, and deployment.

KU Leuven

Sep. 2025 – Jan. 2026

Student Assistant – Object-Oriented Software Development

Leuven, Belgium

- Assisted students during practical programming sessions.
- Guided students through debugging and object-oriented programming exercises.

- Supported teaching staff during laboratory activities.

ARCA Properties

Summer 2023 & Spring 2024

Operations Assistant (Internship)

Luxembourg, Luxembourg

- Diagnosed and resolved maintenance issues across multiple luxury rental properties.
- Assisted with furnishing and commissioning newly acquired apartments.
- Worked directly with the Operations Manager to maintain operational readiness.

INTELLEGO

2022 – 2023

Private Tutor

Echternach, Luxembourg

- Tutored secondary school students in mathematics
- Adapted teaching methods to individual learning needs.

AVENCOR S.A. • COSMOLUX INTERNATIONAL S.A. • Commune de Consdorf

2020 – 2022

Summer Employment

Luxembourg

- Developed teamwork, communication, and problem-solving skills through professional work experience.

Leadership

- Co-Founder of Olympus MARS.
- Organized STEM outreach activities and educational workshops.
- Organized a martial arts workshop during Sport Festival Delux.

Awards & Activities

Awards

- 1st Prize – BEST Leuven Coding Challenge (2025)
- Winner – Ferrero Responsible Marketing Challenge (2023)
- 1st Prize – My First Enterprise (2020)

Competitions

- Engineering Case Competition (VTK)
- Sustainable Skies Challenge (Delta Air Lines)
- Meta Spark Hackathon
- Belgian Mathematical Olympiad – Semi-finalist
- Luxembourg Physics Olympiad – Semi-finalist
- Luxembourg Chemistry Olympiad – Semi-finalist
- Luxembourg Biology Olympiad – Semi-finalist

Professional Development

- BEST Summer Course – Brasov
- First Steps into Consulting Workshop
- Grow Your Future Track
- Engineering Trainee Days – SES ASTRA

Languages

- Portuguese — Native
- English — C1
- French — C1
- Luxembourgish — C1
- German — C1
- Italian — B1